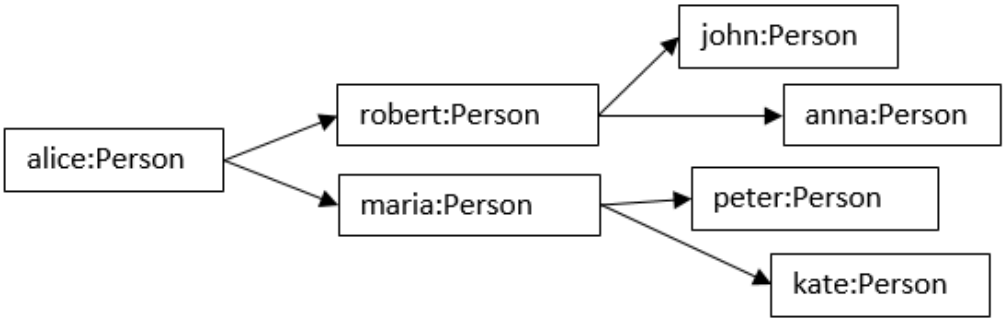
Programming Fundamentals II: Lab 10

Due Date: Tuesday November 19, 2024 at midnight

Part 1. Family Tree Search (10 Marks)

Suppose we have a class called Person. This class has 3 fields called name (String), father (Person) and mother (Person). The person class is said to be *recursive* as it contains references to other persons.

Here is the family tree we will be working with:



a. **(3 Marks)** Write the code for the person class. Include all getters and setters, a full-arg constructor and a constructor that just accepts the name. Also include a setParents method that can set both mother and father.

b. **(1 Mark)** Modify the driver file to create the family tree above using setter methods.

c. **(5 Marks)** Write a method for the person class called checkFamily which accepts a name (String) as input. This method will return true if that person matches the name, or if they have any ancestors that match the name. A person may only have 1 parent. In the event that the person has no known ancestors, return false.

   Hint: You will need to use a recursive algorithm.

   Eg: For this family tree alice.checkFamily("anna") and alice.checkFamily("alice") will return true, but alice.checkFamily("thomas") would return false.

d. **(1 Mark)** Have your Driver to determine the following:
   a. Does alice have an ancestor named anna?
   b. Does alice have an ancestor named alice?
   c. Does alice have an ancestor named thomas?
   d. Does robert have an ancestor named alice?
   e. Does maria have an ancestor named kate?
   f. Does robert have an ancestor named maria?

Add the **date and your name and student number** to the top of your java file. Please submit your java file to Nexus before the end of the day on your lab date. **Late submissions will be not be accepted.**